1. HASH FUNCTION – MD5

AIM

To implement the MD5 hash function and generate the hash value for a given message using Python.

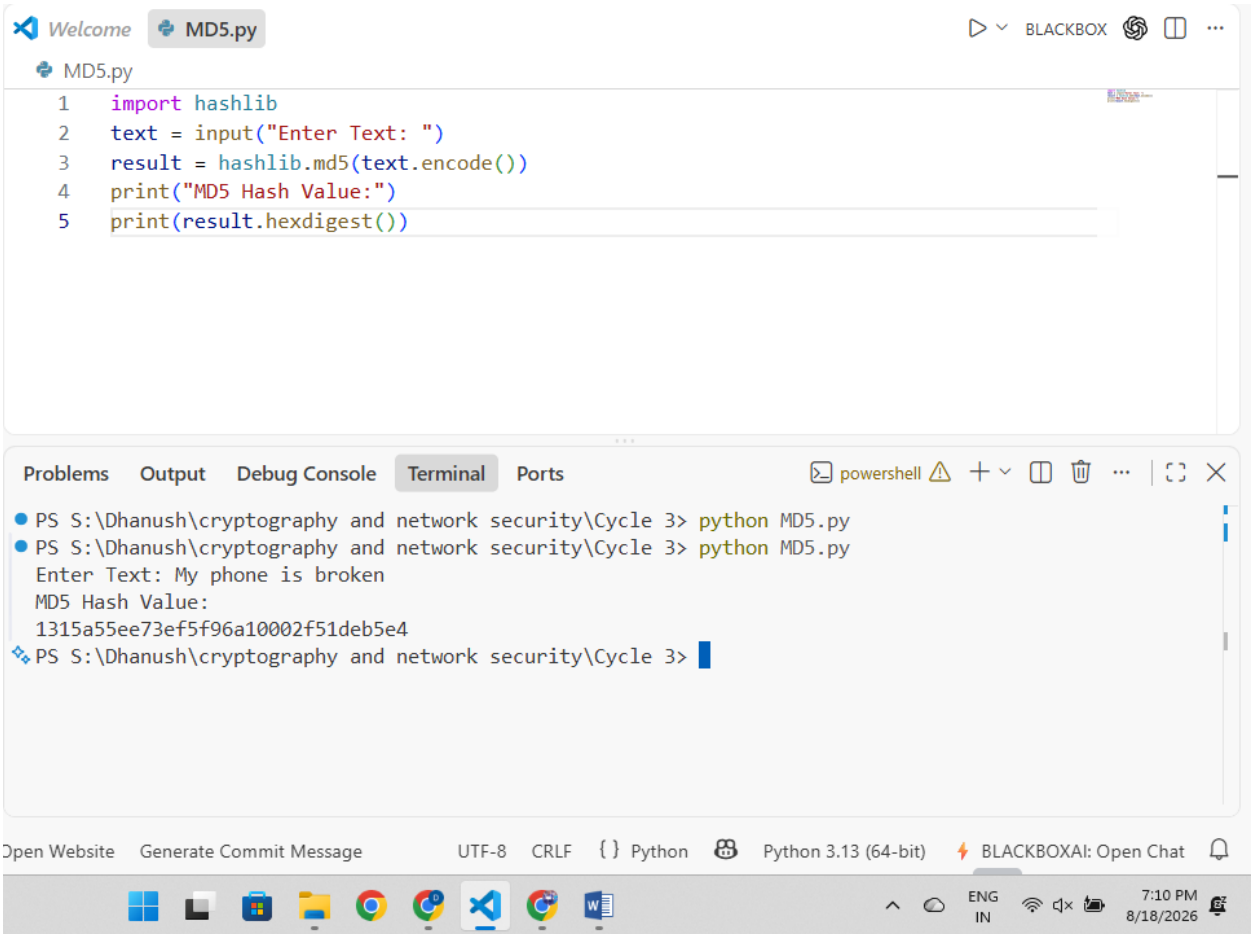
ALGORITHM

1. Start the program.
2. Read the input text from the user.
3. Convert the text into bytes.
4. Apply the MD5 hashing algorithm.
5. Generate the hash value.
6. Display the hash value.
7. Stop the program.

PYTHON CODE

```
import hashlib

text = input("Enter Text: ")

result = hashlib.md5(text.encode())

print("MD5 Hash Value:")
print(result.hexdigest())
```

SAMPLE OUTPUT



RESULT

Thus, the MD5 hash function was implemented successfully and the hash value was generated.

2. HASH FUNCTION – SHA512

AIM

To implement the SHA512 hash function and generate the hash value for a given message using Python.

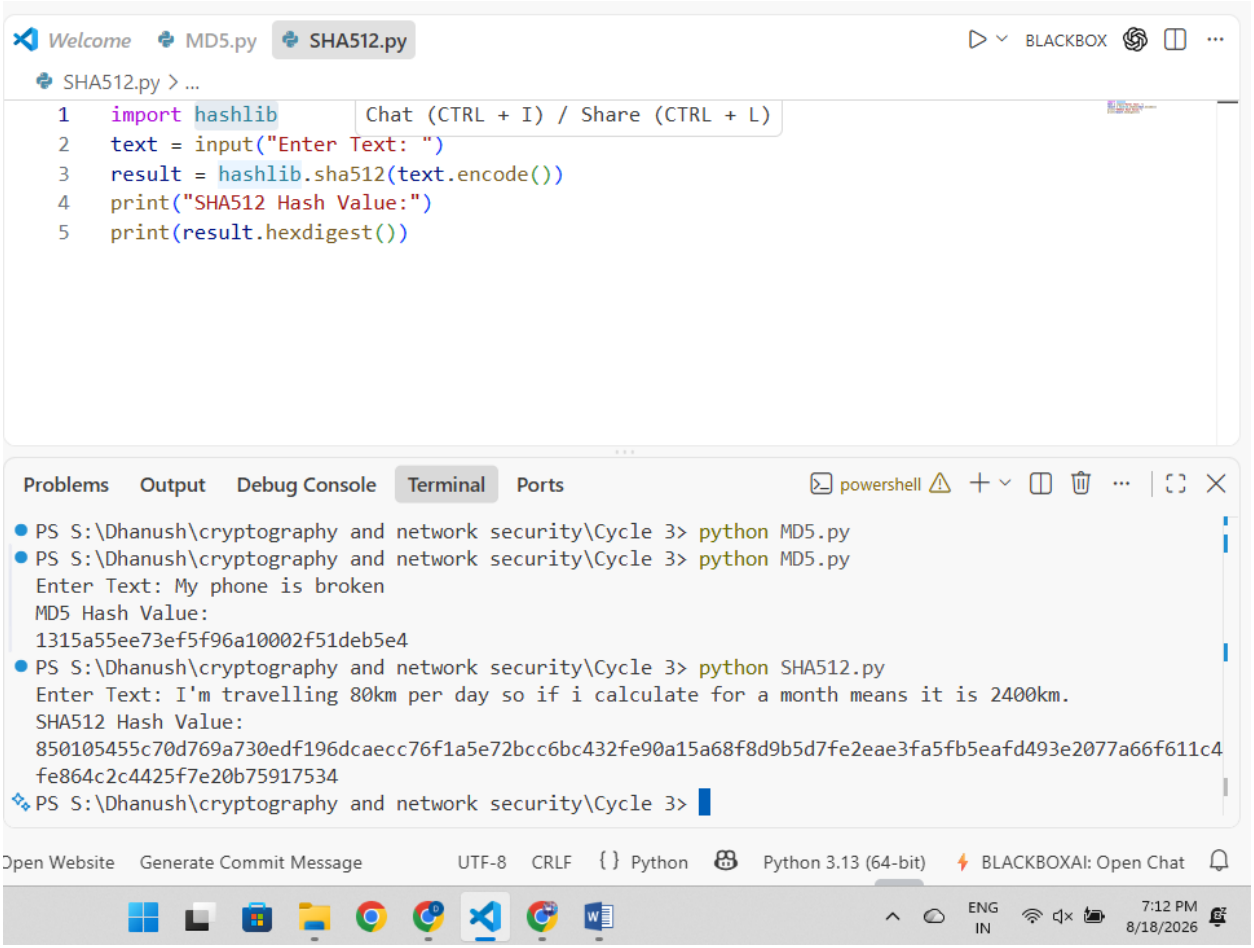
ALGORITHM

1. Start the program.
2. Read the input text from the user.
3. Convert the text into bytes.
4. Apply the SHA512 hashing algorithm.
5. Generate the hash value.
6. Display the hash value.
7. Stop the program.

PYTHON CODE

```
import hashlib

text = input("Enter Text: ")

result = hashlib.sha512(text.encode())

print("SHA512 Hash Value:")
print(result.hexdigest())
```

SAMPLE OUTPUT



RESULT

Thus, the SHA512 hash function was implemented successfully and the hash value was generated.

3. HASH FUNCTION – SHA1

AIM

To implement the SHA1 hash function and generate the hash value for a given message using Python.

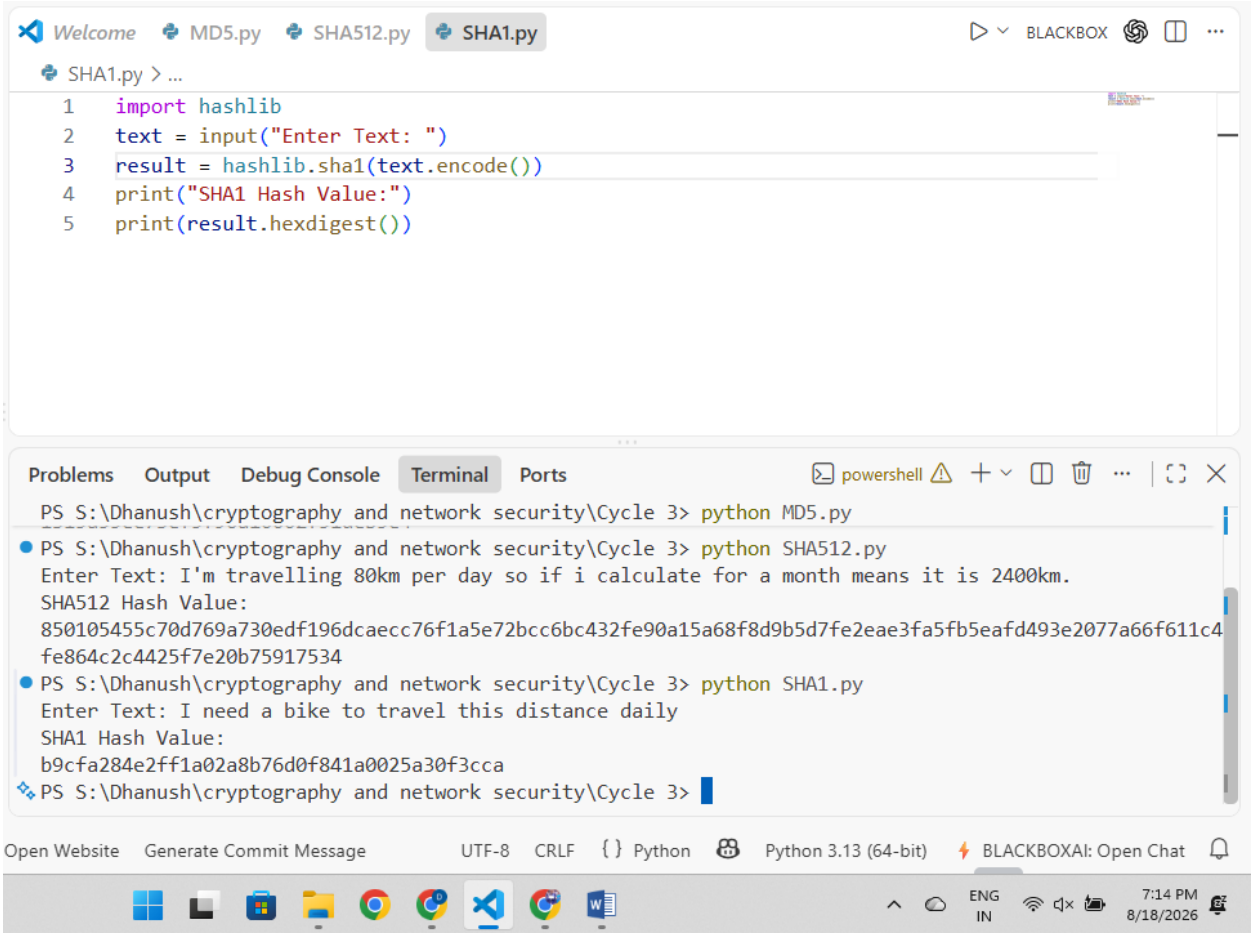
ALGORITHM

1. Start the program.
2. Read the input text from the user.
3. Convert the text into bytes.
4. Apply the SHA1 hashing algorithm.
5. Generate the hash value.
6. Display the hash value.
7. Stop the program.

PYTHON CODE

```
import hashlib

text = input("Enter Text: ")

result = hashlib.sha1(text.encode())

print("SHA1 Hash Value:")
print(result.hexdigest())
```

SAMPLE OUTPUT



RESULT

Thus, the SHA1 hash function was implemented successfully and the hash value was generated.

4. DIGITAL SIGNATURE

AIM

To implement a Digital Signature algorithm for signing and verifying a message using Python.

ALGORITHM

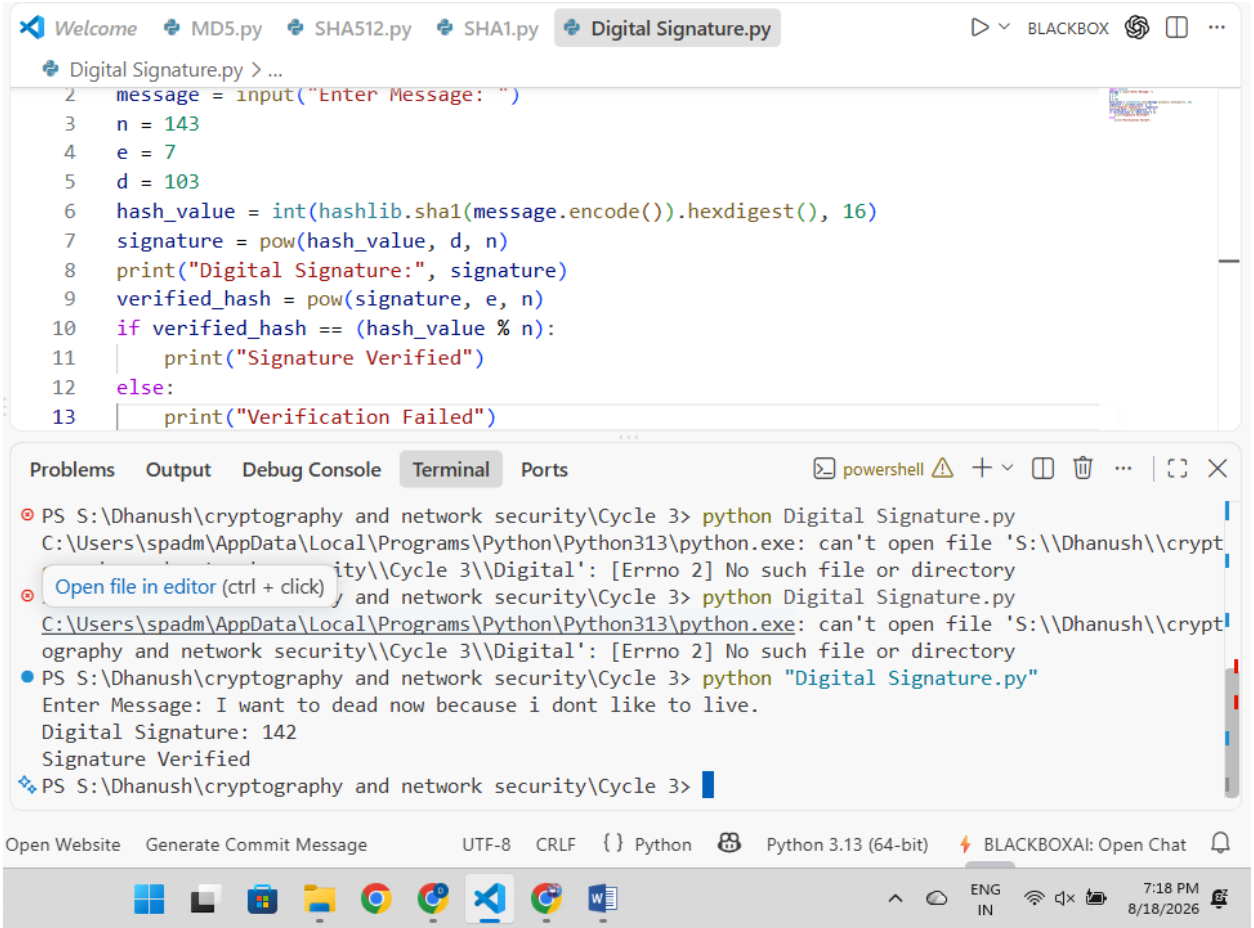
1. Start the program.
2. Read the message from the user.
3. Generate the hash value of the message.
4. Create a digital signature using the private key.
5. Verify the signature using the public key.
6. Display the signature.
7. Display the verification result.
8. Stop the program.

PYTHON CODE

```
import hashlib

message = input("Enter Message: ")

n = 143
e = 7
d = 103

hash_value = int(hashlib.sha1(message.encode()).hexdigest(), 16)

signature = pow(hash_value, d, n)

print("Digital Signature:", signature)
```

```
verified_hash  = pow(signature,  e, n)

if verified_hash  == (hash_value  % n):
    print("Signature  Verified")
else:
    print("Verification  Failed")
```

SAMPLE OUTPUT



RESULT

Thus, the Digital Signature generation and verification process was implemented successfully using Python.